* Time Series - 2

Agenda → Exponential smoothing → family

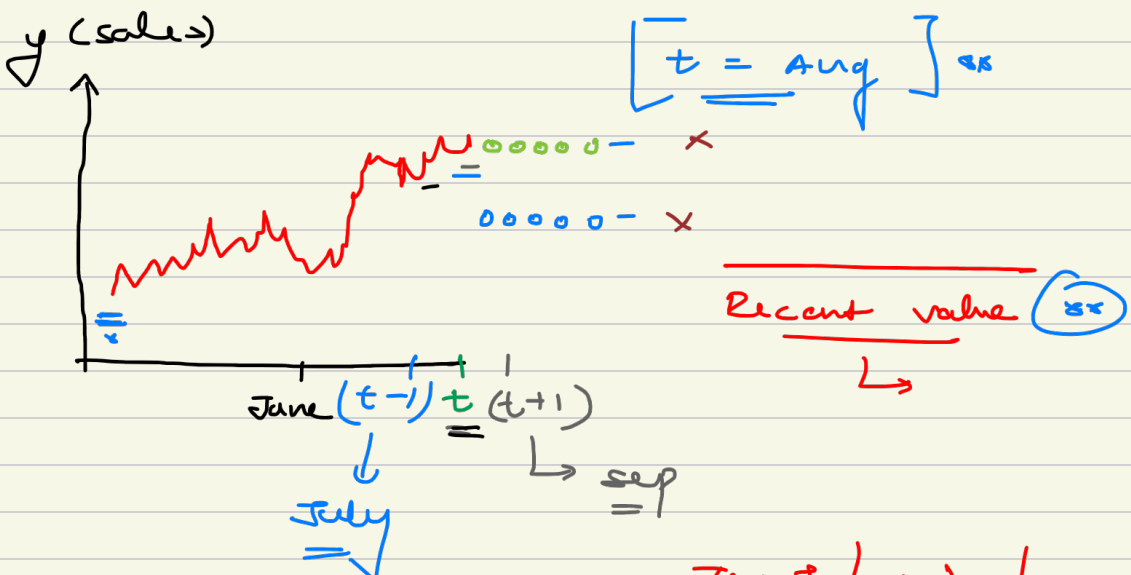
- → simple
- → Double
- → Triple

* →
- mean
- naive
- Drift

- Stationarity | ACF

** Simple Expo. Smoothing [SES]

Naive + mean + window



TRUST / wt. / confidence

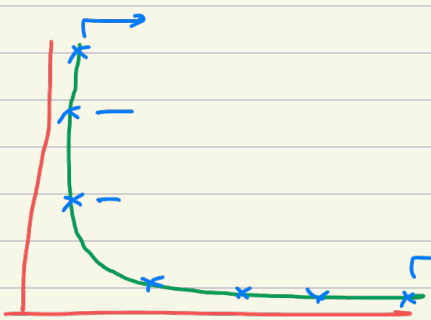
No equal weightage

⇒ wts ↓ as u go past in time

(t+1) Highest

$$\hat{y}_{t+1} = w_1 y_t + w_2 y_{t-1} + w_3 y_{t-2} + \dots + y_0$$

wts ? → Exponential smoothing



↳ Recent values ✓

↓

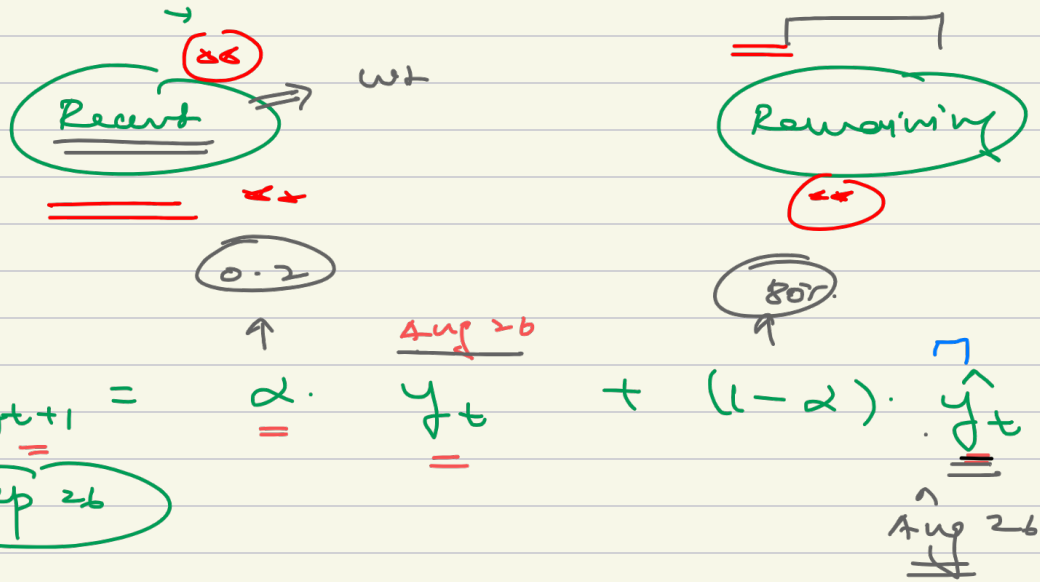
Trend ✗✗

↳ $w \simeq 0.0001$ ↓

Reduce wts

Very quickly

✗



* Recursive formulation

$$\hat{y}_{t+1} = \alpha \cdot y_t + (1-\alpha)\left[\alpha \cdot y_{t-1} + (1-\alpha)\hat{y}_{t-1}\right]$$

Sep → Aug → July → June → May

$\hat{y}_1 = 0$ or ...

$$\hat{y}_{t+1} = \alpha \cdot y_t + \alpha(1-\alpha) y_{t-1} + \alpha(1-\alpha)^2 y_{t-2} + \alpha(1-\alpha)^3 y_{t-3} + \cdots$$

$\alpha \in (0,1)$

80% → Avg

0.8×0.2

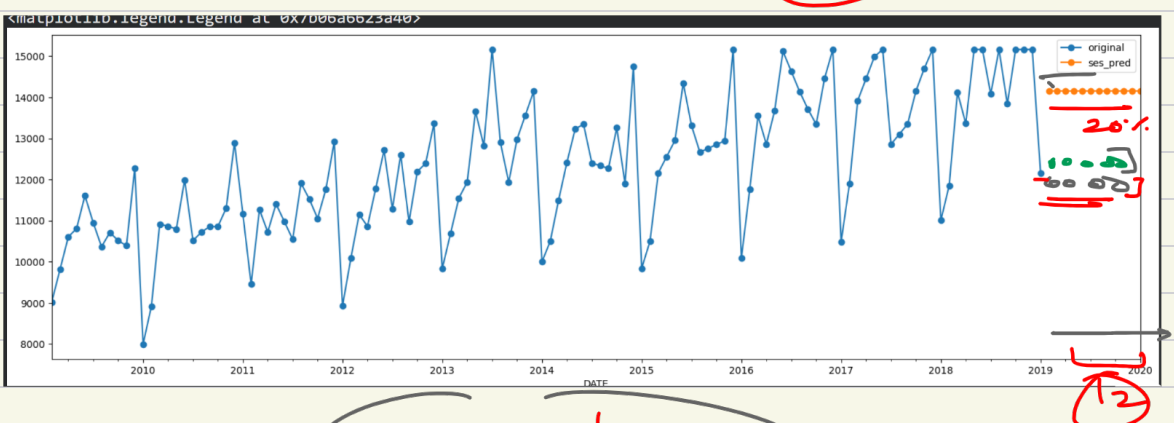
$[0.8 \times 0.2 \times 0.2]$

$$\hat{y}_{t+1} = 0.8 y_t + 0.16 y_{t-1} + 0.032 y_{t-2}$$

$\Rightarrow y = 0$

TS stop

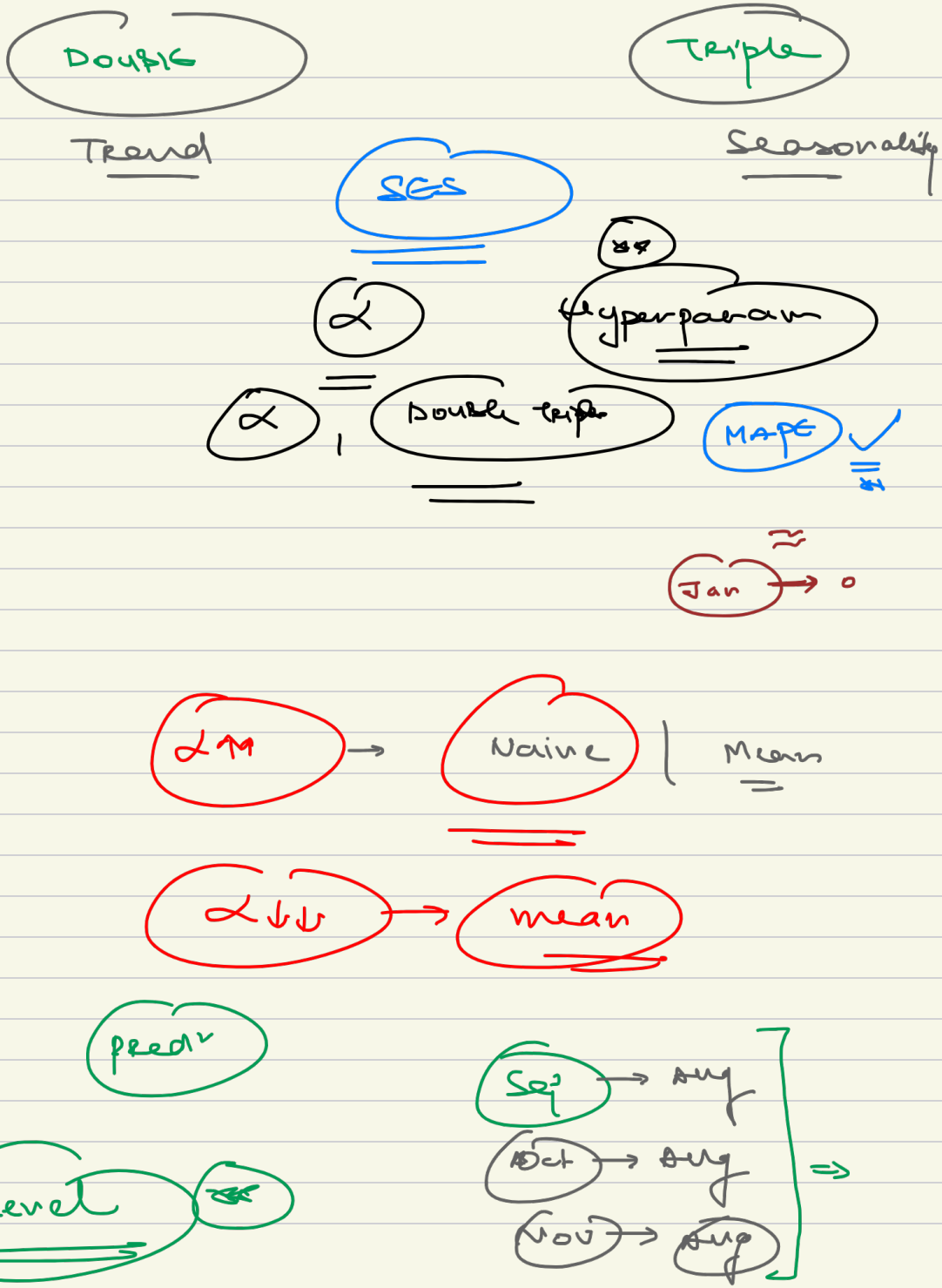
α [SES → current level of α]



$\alpha = 1$ / 100% → Naive approach

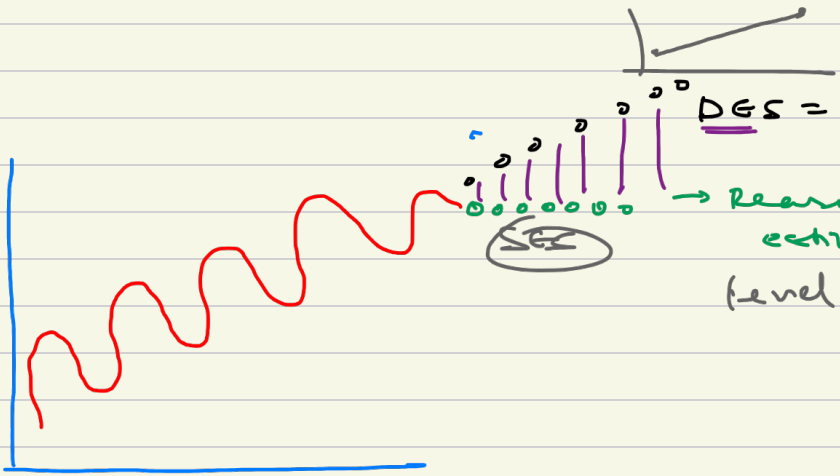
SES → Reasonable estimate of TS

(*) Naive / Avg


Double
Triple
Trend
Seasonality
SES
α
hyperparam
α
Double triple
MAPE
Jan
0
α M
Naive
Mean
α ↓↓↓
mean
pred^n
Level
Seq
Oct
Nov
Avg
Avg
Avg
⇒

DES — DOUBLE Exponential smoothing

SES + Drift method → Trend



Drift ⇒

DES

Business

Trend ?

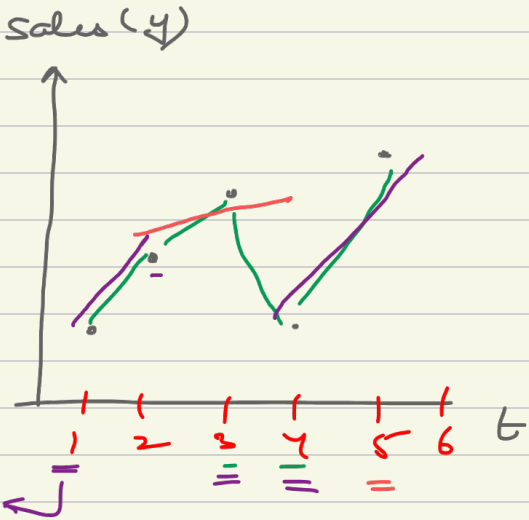


growth ?

Trend = slope

Slope = Rate of change

$$\text{Slope} = \frac{\Delta y}{\Delta t} = \frac{y_2 - y_1}{t_2 - t_1}$$



$y'(t)$ = slope at timestamp t

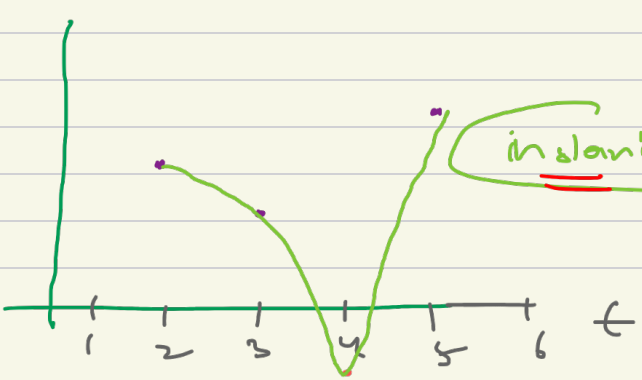
= Trend

= instantaneous Trend

$$\frac{\Delta y}{\Delta t} = \frac{y_2 - y_1}{t_2 - t_1}$$

$$= 1$$

$$y_2 - y_1$$



instantaneous Trend

SES

SEC

(1) SES → Reasonable level of y

(2) estimate of Trend

→ take avg

→ latest Trend

Combⁿ of Both → SES

Some wt → most instantaneous Trend

(3) Remaining → Trend that may been going for many months

(4) Expo smoothing

TS (sales) → Expo → SES → level

(5) Δy → Trend / slope

DES → Trend

Reasonable estimate of Trend

DES = SES + Trend

mathematics

$\hat{y}_{t+h}$

Δy

$\hat{y}_{t+h} = l_t + h \cdot b_t$

l_t : avg

b_t : avg → Trend

l_t : Sep → CURRENT level → SES

$\hat{y}_{t+1} = l_t + 1 \cdot b_t$ ⇒ SES

Sep

$\hat{y}_{t+2} = l_t + 2 \cdot b_t$ → Trend

SES eqn

$$l_t = \alpha \cdot y_t + (1-\alpha)\left[l_{t-1} + b_{t-1}\right]$$

level — l_t — y_t — SES

Reasonable trend

$$b_t = \beta\left[l_t - l_{t-1}\right] + (1-\beta)\hat{b}_{t-1}$$

$l_t - l_{t-1}$ → current trend

$\hat{b}_{t-1}$ → previous trend

2 hyperparams

α → level

β → trend

$\alpha = 1$ → naive

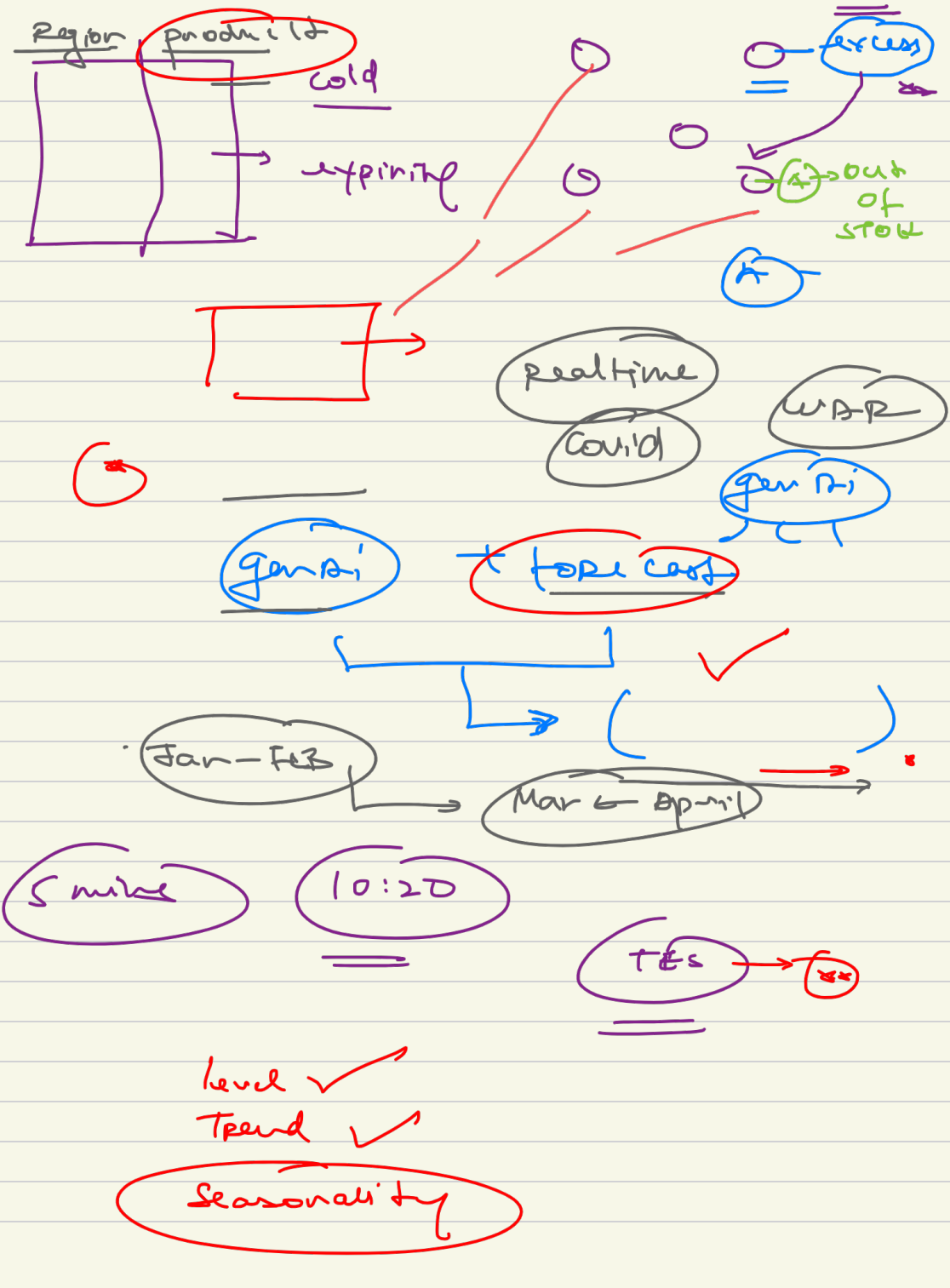
$\alpha = 0$

Holt

Holt. avg

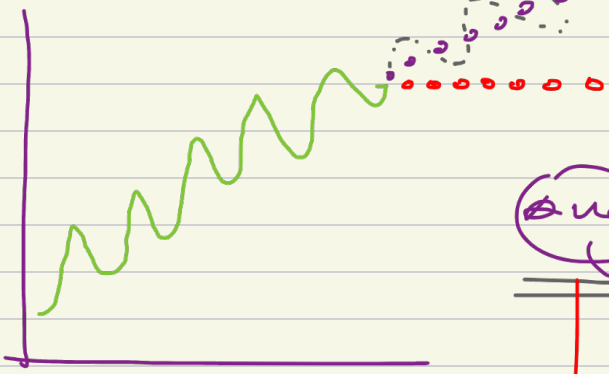
Overall trend

Inventory → MacBook pro M1


Region
product id
cold
expiring
excess
out of stock
realtime
covid
WAR
gen AI
gen AI
forecast
Jan - Feb
Mar - April
Smile
10:20
TES
level
Trend
Seasonality

TES = DES + seasonality

DES = SES + Trend



26

Aug

Aug 25

instantaneous

Records

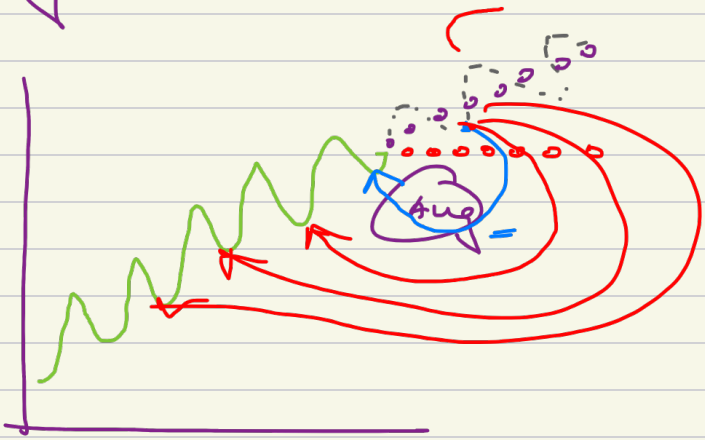
Aug 25 + Aug 24

+ Aug 23

—:—

Aug

Aug data



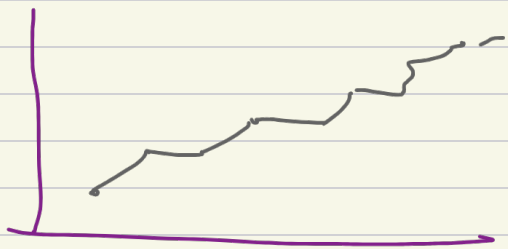
20 yrs

exponential smoothing

wt avg → Aug seasonality

Aug DATA → Exp smoothing → Seasonal components

Aug



Dec

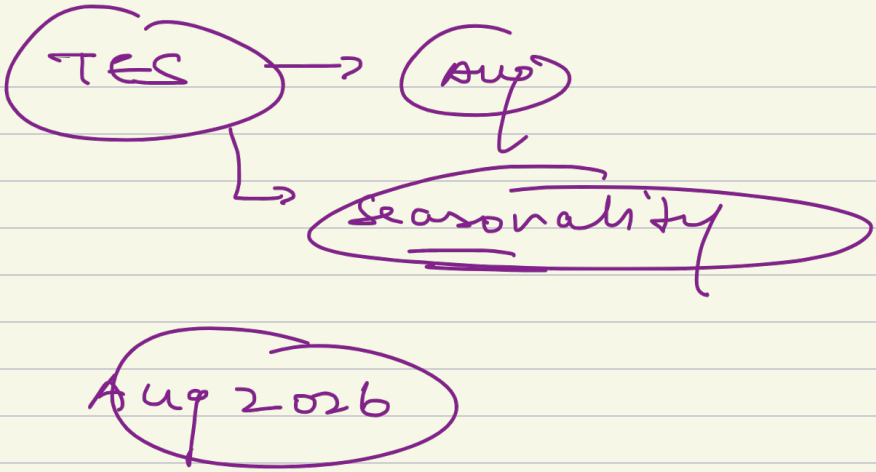


Reasonable estimates of seasonality

SES → current level

DES → Recent + past Trends

→ Trend



$$\text{Aug 2026} = DES + \gamma(\text{Aug 25}) + (1-\gamma)\widehat{\text{Aug 25}}$$

$$= DES + \gamma\,\text{Aug 25} + \gamma(1-\gamma)\,\text{Aug 24} + \gamma(1-\gamma)^2\,\text{Aug 23} + \cdots$$

γ = Hyperparam

α

β

Trend

$\hat{y}_{t+h} = l_t + h \cdot b_t + S_{t+h-M}$

l_t → current level

$M = 12$

$t+1$

Sep 2026

$l_t + 1 \cdot b_t + S_{t+1-12}$

l_t → Avg level

b_t → Trend

S_{t+1-12} → Avg Sep 2T

TES

Holt winters forecasting

Quiz time!

Time Left: 32s

Which model is more appropriate for forecasting a time series with both trend and seasonal components?

20 users have participated

A	SES (Single Exponential Smoothing)	0%
B	DES (Double Exponential Smoothing)	20%
✓ C	TES (Triple Exponential Smoothing)	75%
D	Moving Average	5%

End Quiz Now

ponential smoothing

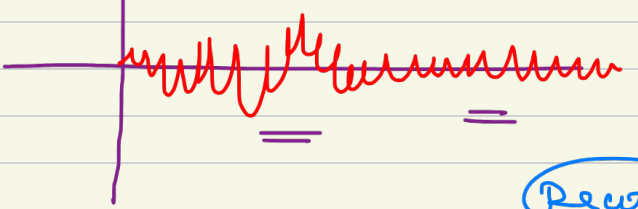
ARIMA →

ST ↳ =

y

X No trend

X No seasonality



Residuals

Prophet → trend, seasonality, error

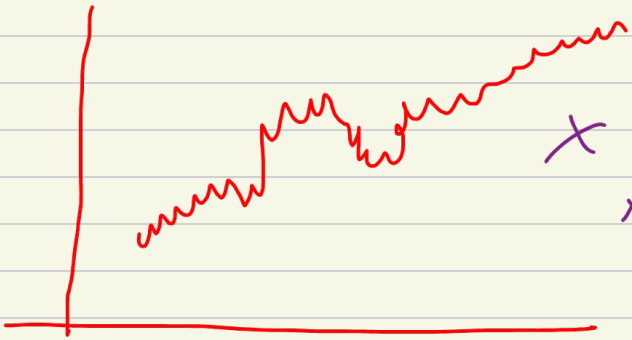
$$y(t) = b(t) + s(t) + e(t)$$

ARIMA → TS → Stationarity

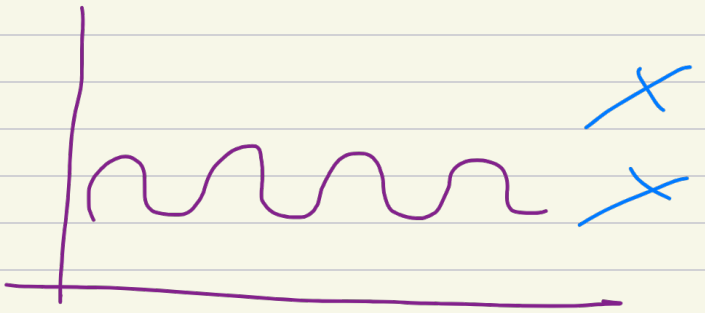
Stationarity → params / properties

μ, σ^2, amplitude, frequency, seasonality

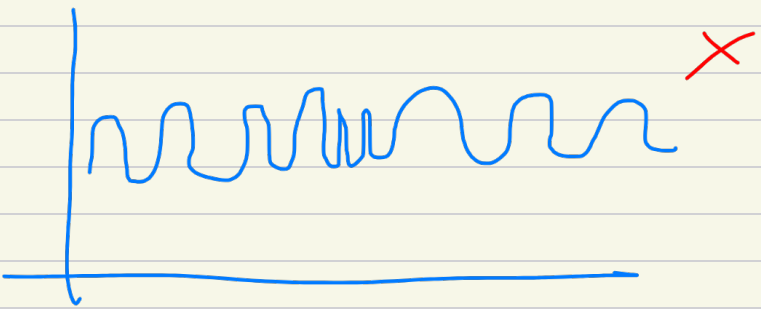
X



Non stationary

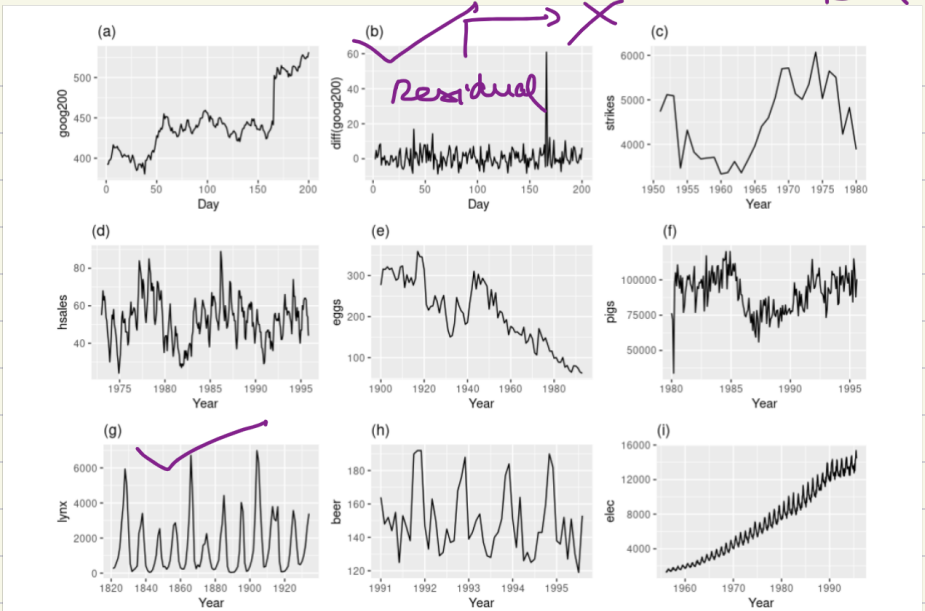


Seasonal



freq. changing

Real world



Stationarity

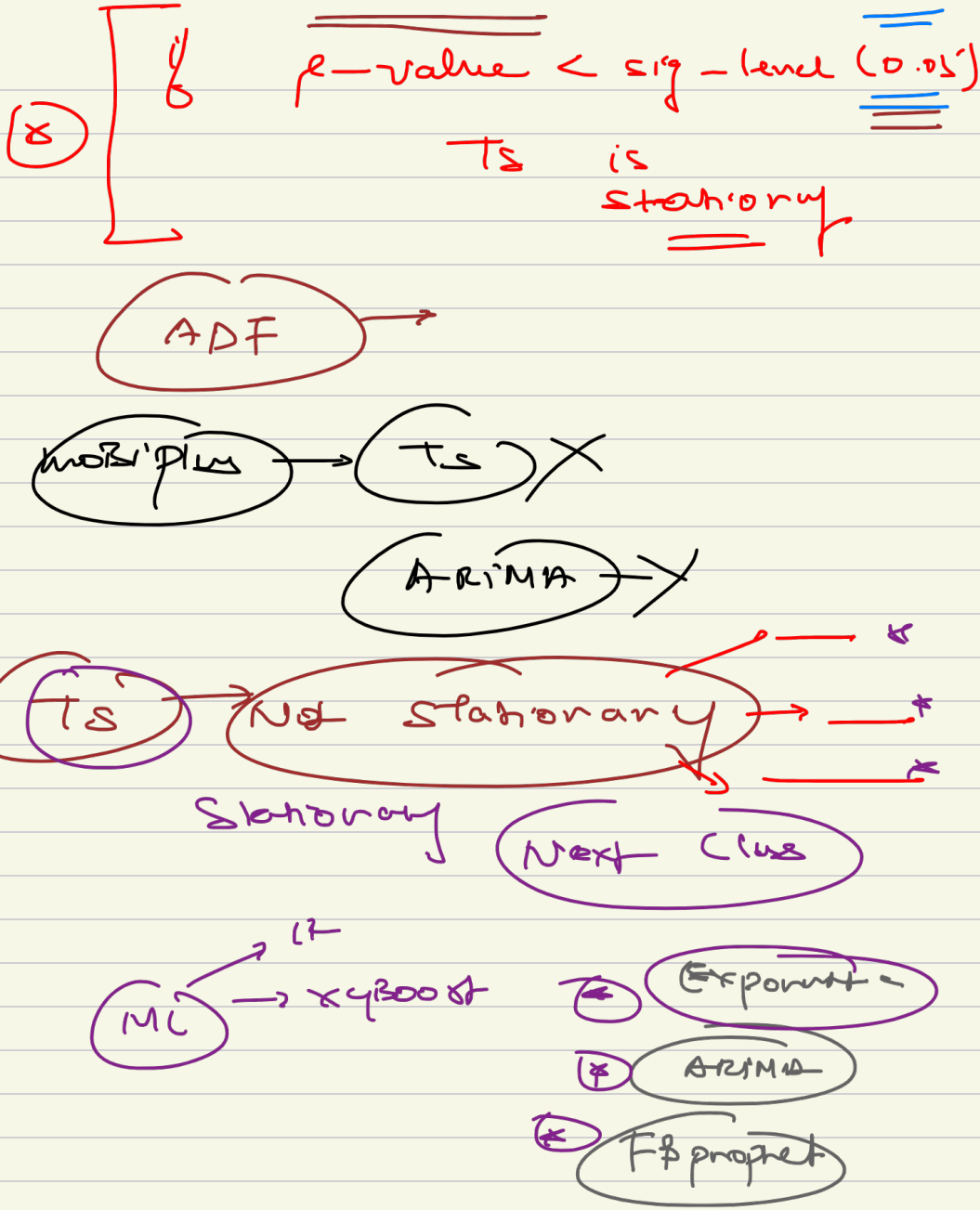
TS → Stationary or not

DICKEY FULLER TEST

H_0: TS is Non Stationary

H_1: TS is Stationary

evidence


p-value < sig-level (0.05)
Ts is stationary
ADF
Multiplicative
Ts
ARIMA
Ts
Not Stationary
Stationary
Next Class
ML
XGBoost
Exponential
ARIMA
FB prophet

Quiz time!

Time Left: 43s

What is the purpose of the Dickey-Fuller test in time series analysis?

17 users have participated

A	To confirm the presence of a linear trend.	6%
B	To determine the seasonality of a time series.	12%
C	To calculate the future values of the series.	0%
✓ D	To test for stationarity in a time series.	82%

End Quiz Now

